

## Old UNM Algebra Qualifier Problems

Based on Chapter 4 of Dummit–Foote

Fifteen-year collection: January 2012–August 2023

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**Scope.** This packet uses the third edition of Dummit and Foote, *Abstract Algebra*. Chapter 4 is “Group Actions”. The first section contains problems that can reasonably be attempted after this chapter. The second section records closely related problems.

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### I. Core problems from Chapter 4

#### January 2012 Qualifier, Problem 1

How many elements of order 7 must be in a simple group of order 168?

#### January 2013 Qualifier, Problem 1

Let  $p$  be a prime. Show that every group with  $p^2$  elements is abelian. Give an example of a nonabelian group with  $p^3$  elements.

#### January 2017 Qualifier, Problem 1

Suppose  $p$  and  $q$  are distinct primes such that

$$p \nmid q \pm 1 \quad \text{and} \quad q \nmid p \pm 1,$$

and let  $G$  be a group of order  $p^2q^2$ . Show that  $G$  is abelian.

#### January 2017 Qualifier, Problem 2

Determine the number of Sylow  $p$ -subgroups in  $S_p$ .

#### August 2017 Qualifier, Problem 3

Suppose  $p$  and  $q$  are distinct primes. Show that if

$$p \nmid q - 1 \quad \text{and} \quad q \nmid p \pm 1,$$

then every group of order  $p^2q$  is abelian.

#### January 2018 Qualifier, Problem 8

Let  $p$  be a prime, let  $G$  be a finite group whose order is divisible by  $p$ , and let  $P$  be a normal  $p$ -subgroup of  $G$ . Show that  $P$  is contained in every Sylow  $p$ -subgroup of  $G$ .

#### August 2019 Qualifier, Problem 5

Let  $p, q, r$  be primes with  $p < q < r$ . Show that a group  $G$  of order  $pqr$  has a normal Sylow subgroup of order  $p$ ,  $q$ , or  $r$ .

#### January 2020 Qualifier, Problem 3

What is the order of a Sylow  $p$ -subgroup of  $S_{3p}$  for  $p > 3$ ? Are the Sylow  $p$ -subgroups of  $S_{3p}$  abelian? Justify your answers.

**August 2020 Qualifier, Problem 2**

Consider  $G = S_{28}$ .

- (a) What is the order of a Sylow 7-subgroup of  $G$ ?
- (b) Is a Sylow 7-subgroup abelian? Justify your answer.

## II. Closely related later-material problem

**January 2018 Qualifier, Problem 9**

Let  $p$  and  $q$  be distinct primes and let  $G$  be a group of order  $pq^r$  for some  $r \geq 1$ . If

$$q \nmid p-1 \quad \text{and} \quad p \nmid q^i-1 \quad \text{for every } 1 \leq i \leq r,$$

show that  $G$  is nilpotent. For what values of  $r$  must  $G$  be abelian?

### Source

University of New Mexico, Department of Mathematics and Statistics, [Past Qualifying Exams – Algebra](#). Dates and problem numbers are recorded above.